

Lecture 2: Types of Data, Data Collection & Sampling Methods

Course: Introduction to Data Analytics

Understanding Types • Collection Process • Primary vs Secondary • Sampling Introduction

Dr . Mohammed Khalaf

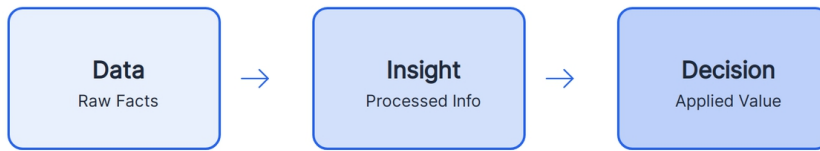
Recap of Lecture 1



- **Definition of Analytics:** Discovering meaningful patterns from raw data.
- **The 4 Types:** Descriptive, Diagnostic, Predictive, and Prescriptive.
- **The Lifecycle:** Collection, Cleansing, Analysis, and Visualization.
- **Today's Focus: Data Collection**

What is Data?

Data is the raw form of information, consisting of **facts, figures, and symbols**. In its simplest state, data is disorganized and lacks context.



Data becomes valuable only when it is organized, interpreted, and applied.

The Role of Data

1. Decision-making

Empowers leaders to move beyond intuition, making choices based on evidence and historical patterns to drive better outcomes.

2. AI/ML Innovation

Data acts as the "fuel" for Artificial Intelligence and Machine Learning models, enabling systems to learn, adapt, and predict.

3. Digital Transformation

Facilitates real-time analytics and organizational agility, allowing businesses to respond instantly to market changes.



Types of Data

1. Quantitative Data

Definition: Measurable, counted, and expressed in numerical terms. Ideal for statistical analysis.

Sub-types:

- **Discrete:** Whole numbers that can be counted (e.g., number of employees).
- **Continuous:** Data on a scale that can be measured (e.g., temperature, revenue).

Examples: Age, Weight, Market Price, Sales Figures



Activity: Identify the Data Type



Item	Data Type
1. Favorite color	Qualitative (Categorical)
2. Number of emails received	Quantitative (Discrete)
3. File download time	Quantitative (Continuous)
4. Country of birth	Qualitative (Categorical)
5. Hourly temperature	Quantitative (Continuous)

2. Qualitative Data

Definition: Non-numeric, descriptive information that focuses on qualities and characteristics.

Core Focus:

- **The "Why":** Captures opinions, motivations, and deep insights.
- **Categories:** Groups data by shared traits rather than numbers.



Examples: Customer Feedback ("satisfied"), Color, Audio Recordings, Open-ended Comments

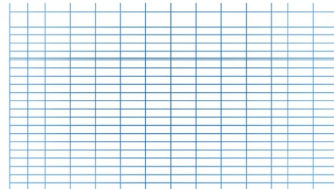
3. Structured Data

Definition: Data that resides in a fixed field within a record or file, typically in rows and columns.

Key Traits:

- **Organized:** Stored in Relational Databases (SQL).
- **Searchable:** High accuracy and easy to query.

Examples: Bank Transactions, Inventory Lists, Excel Spreadsheets, Flight Bookings



4. Unstructured Data

Definition: Information that does not have a predefined data model or is not organized in a pre-defined manner.

The Scale:

Unstructured data accounts for **>80%** of all global data generated today.

Examples: Emails, Video/Audio Files, PDFs, Social Media Posts, Images



5. Semi-Structured Data

Definition: A hybrid form of data that does not reside in a relational database but contains tags or markers to separate elements.

Key Traits:

- **Hybrid:** Combines flexibility with a degree of organization.
- **Formats:** Commonly found in JSON, XML, and NoSQL databases.

Examples: Sensor Logs, Web Data (HTML), Email Headers, JSON APIs



Big Data & Characteristics

What is Big Data?

Big Data refers to datasets that are **too large, too fast, or too complex** to be managed, stored, or analyzed using traditional data processing methods.

The Turning Point:

Big Data begins where traditional database management systems (DBMS) and desktop processing tools fail to provide timely results.

The Challenge:

It's not just about size; it's about the difficulty of *capturing, storing, searching, sharing, and analyzing* data at such massive scales.

The 5 Vs of Big Data



Volume

The sheer amount of data generated every second.



Velocity

The speed at which new data is generated and moves.



Variety

The different formats (text, video, sensor data).



Veracity

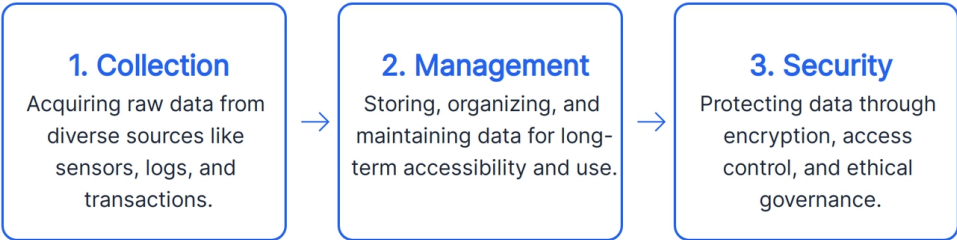
The accuracy, quality, and trustworthiness of data.



Value

The ability to turn data into meaningful insights.

Data Lifecycle



Applications of Data

Industry	Key Use Case
Business	Predicting consumer trends and optimizing supply chains.
Healthcare	Precision medicine and faster, data-driven disease diagnosis.
Finance	Real-time fraud detection and personalized financial advice.
Tech	Fueling AI models, NLP, and automated decision systems.
Marketing	Hyper-personalization and targeted advertising campaigns.

Key Challenges

Quality

"Garbage in, garbage out." Poor data quality leads to unreliable insights and bad decisions.

Volume

Managing and storing massive scales of data requires significant infrastructure and cost.

Privacy & Ethics

Navigating GDPR, data protection laws, and the ethical use of personal information.

Bias

Implicit biases in data can lead to "unfair AI," reinforcing discrimination in automated systems.

Effort & Cost: The time and financial investment required for data cleaning, processing, and expert analysis is often underestimated.

Learning Objectives

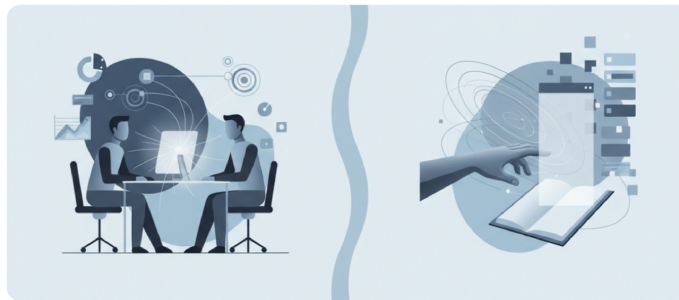
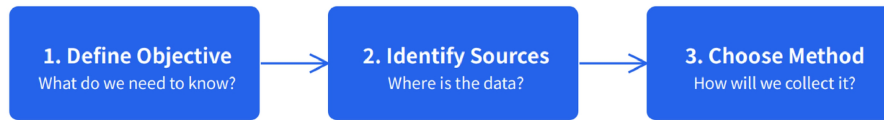


By the end of this lecture, you will be able to:

Distinguish between data types: Qualitative vs. Quantitative and Discrete vs.

1. Continuous.
2. Explain the difference between Primary and Secondary data sources.
3. Define Population and Sample and understand the relationship between them.
4. Identify and compare various Sampling methods and their applications.

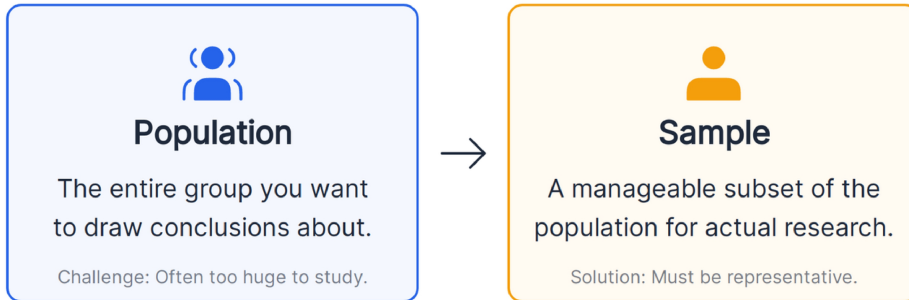
The Data Collection Process



Primary vs. Secondary Data

	Primary Data	Secondary Data
Source	You collect it first-hand for your specific purpose.	Collected by others for a different purpose.
Advantages	• High control over quality • Tailored to your objectives	• Time and cost efficient • Easy to access
Disadvantages	• Expensive and time-consuming • Requires expertise	• May not fit your specific needs • Reliability might be unknown
Examples	Surveys, Experiments, Interviews.	Census reports, Journals, Govt websites.

Populations and Samples



We use **Samples** to save *time* and *money* while maintaining accuracy.

Visualizing Population and Sample



Key Takeaway: We study the **SAMPLE** to make an **INFERENCE** about the **POPULATION**.

Sampling Methods: Two Categories



Probability Sampling

Every member has a known, equal chance of selection.

"The Gold Standard"



Non-Probability Sampling

Selection based on non-random criteria (e.g., convenience).

Higher Risk of Bias



Probability Sampling Methods



Simple Random

Completely random selection.
(e.g., Names in a hat)

Best for homogeneous groups.



Stratified Random

Dividing population into groups (strata) then sampling.
(e.g., By grade level)



Cluster Sampling

Randomly selecting entire groups or locations.
(e.g., Neighborhoods)

These methods ensure every individual has an **equal chance** of being included, minimizing bias.

Non-Probability Sampling Methods

- **Convenience Sampling:** Collecting data from people who are easiest to reach (e.g., classmates, friends).
- **Purposive / Judgmental Sampling:** The researcher uses their expertise to select a sample most useful to the objective.
- **Snowball Sampling:** Existing participants recruit further participants from among their acquaintances.



Warning: Use with Caution

These methods are prone to **high bias** and may not represent the population accurately.

The Danger of Sampling Bias

Bias: A systematic error that skews results in one direction, making the sample unrepresentative.

Case Study: 1936 Literary Digest Poll

- **The Survey:** 10 million surveys sent, 2.4 million responses.
- **The Result:** Wrongly predicted the presidential election winner.
- **The Flaw:** They sampled car and telephone owners—a wealthy demographic during the Depression.

CRITICAL LESSON: Sample Size DOES NOT Fix Bias!

Summary & Key Takeaways

- **Data Types:** Understand Qual vs Quan and Discrete vs Continuous.
- **Sources:** Primary (Tailored) vs Secondary (Existing).
- **Pop vs Sample:** We use samples to infer about populations.
- **Sampling:** Probability (Random) vs Non-Probability (Judgmental).
- **Bias is the Enemy:** Size is no substitute for representativeness.

Questions?

Thank you for participating in Lecture 2!